

VRP Options Strategy

Strategy Explanation & Look-Ahead Bias Audit

This document provides two levels of explanation for the systematic SPX options strategy implemented in the backtesting codebase, followed by a thorough audit of potential look-ahead bias, with particular attention to COVID-period avoidance.

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Part 1: The Simple Explanation

What does this strategy actually do?

Imagine you run an insurance company, but instead of insuring houses or cars, you insure investors against the stock market crashing. Investors who own stocks are scared of big drops, so they pay you a fee (called a *premium*) for protection. If the market stays calm, you keep all the money they paid you. If the market crashes, you have to pay them back, and it can cost you a lot.

The clever part is that investors tend to **overpay** for this insurance. They are more scared of crashes than they should be, so the price of protection is usually higher than the actual risk. This gap between what they pay and what the risk is really worth is called the **Variance Risk Premium (VRP)**. It is the edge that makes this strategy profitable over time.

How does the strategy decide when to trade?

The strategy uses two traffic lights that both need to be green before it sells insurance:

Traffic Light 1 (VRP Signal): "Is insurance overpriced right now?" A machine learning model predicts how volatile the market will actually be, and compares that to what investors are paying for protection. If the gap is big (insurance is expensive relative to actual risk), the light is green. If the gap is tiny or negative (insurance is cheap or underpriced), the light is red.

Traffic Light 2 (Regime Signal): "How confident is our crystal ball?" The model also says how uncertain it is about its own prediction. If the model is confident (narrow range of possible outcomes), the light is green. If the model is uncertain (wide range), the light turns yellow or red. This stops us selling insurance when we really do not know what is coming.

Only when both lights are green does the strategy enter a trade. The brighter the green on each light, the bigger the trade. There is also a circuit breaker: if the strategy loses too much in one month, it stops trading for a cooling-off period, just like a fuse blowing to prevent a fire.

What does the actual trade look like?

The strategy sells **bull put spreads** on the S&P 500 index. In plain terms: it sells expensive insurance (a put option at a strike price somewhat below the current market level) and buys cheaper insurance further away as a safety net. The difference in price is the profit if the market stays above the short strike. The safety net limits how much you can lose if things go badly wrong.

Each trade has a defined maximum loss, and the strategy sizes positions so that no single trade can wipe out more than a set percentage of the portfolio.

Part 2: Graduate-Level Explanation

Theoretical Foundation

The strategy harvests the Variance Risk Premium (VRP), defined as the spread between implied variance (proxied by VIX^2) and realised variance. The VRP is persistently positive due to investors' concavity in utility of wealth and the resulting demand for downside protection, which inflates the risk-neutral expectation of variance above the physical measure. This premium is the theoretical edge.

Signal Architecture

Signal 1 (VRP Level): The system computes forward-looking VRP as:

$$VRP(t) = VIX(t) - RV_{predicted}(t+h)$$

where $RV_{predicted}$ is the next-period realised volatility forecast from an XGBoost regression model trained on ~150 features. The raw VRP is standardised via a 252-day rolling z-score, and mapped to a discrete regime classification:

z-score	Regime	Scalar
$z > 1.5$	EXTREMELY_RICH	1.00 (+ call overlay eligible)
$0.5 < z < 1.5$	RICH	1.00
$0 < z < 0.5$	NORMAL	0.67
$-0.5 < z < 0$	LEAN	0.33
$z < -0.5$	INVERTED	0.00 (no trade)

A hard floor requires $VRP_{raw} > 0$; negative VRP implies no compensation for bearing variance risk and the strategy stands aside.

Signal 2 (Forecast Uncertainty Regime): Computed from the prediction interval width of the VIX regression model:

$$PI_{width}(t) = PI_{upper_95}(t) - PI_{lower_95}(t)$$

This is z-scored over a 252-day rolling window. Wide prediction intervals imply model uncertainty (fat-tailed regime), which is mapped to a multiplicative dampener:

z-score	Regime	Multiplier
$z < 0.5$	STABLE	1.00
$0.5 < z < 1.0$	ELEVATED	0.75
$1.0 < z < 1.5$	STRESSED	0.50
$z > 1.5$	CRISIS	0.00 (no trade)

Composite Signal and Position Sizing

The two signals are AND-gated: both must permit trading. The composite size scalar is:

$$raw_scalar = S1_scalar * S2_multiplier$$

$$final_scalar = \sqrt{1 - raw_scalar}$$

Note: This formula appears inverted (see audit, Part 3). When both signals are maximally favourable (raw_scalar = 1.0), the final scalar is 0.0, producing zero position size. This is almost certainly a bug.

Supplementary classifiers (VIX direction, RV direction) apply multiplicative reductions when their predicted probability of an adverse move exceeds configurable thresholds. The final scalar determines position size as a fraction of the risk budget.

Trade Construction

The strategy constructs **bull put spreads** on SPX options. The short put is selected nearest to a target delta (default ~16-delta, roughly 1 standard deviation OTM). The long put is placed to create the widest spread affordable within the per-trade risk budget ($\text{NAV} * \text{max_risk_pct} * \text{final_scalar}$). Slippage is modelled as a fraction of the bid-ask half-spread. Positions are held to expiry with no active management.

At expiry, the settlement P&L for each spread leg is computed analytically. An optional naked call overlay is sold when VRP is EXTREMELY_RICH and regime is STABLE, adding upside premium at the cost of unlimited upside risk.

Risk Management

Per-trade risk is capped at max_risk_pct_nav of current NAV. Margin utilisation is bounded by max_margin_utilisation. A monthly circuit breaker triggers a configurable cool-off period if cumulative monthly P&L exceeds a drawdown threshold. Only one position per expiry cycle is permitted.

Part 3: Look-Ahead Bias Audit

The strategy reportedly avoided trading during the COVID crash (Feb-Mar 2020), which preserved capital. While this *could* reflect genuine signal quality, several structural features of the codebase create a high risk of look-ahead bias that could explain this result. Below is a systematic audit.

Summary of Findings

Issue	Severity	Detail
XGBoost training window	CRITICAL	If the RV prediction model was trained on data including COVID, it would learn to predict high RV in the lead-up to COVID, making VRP negative and blocking trades. This is the most likely explanation for COVID avoidance.
VIX regression PI calibration	CRITICAL	Same issue for Signal 2. If the regression model saw COVID data, prediction intervals widen during pre-COVID periods with similar features, triggering CRISIS regime.
Classification model leakage	CRITICAL	Supplementary VIX/RV classifiers trained on full data would learn to flag pre-crash periods, reducing position sizes ahead of the crash.
Composite scalar formula	HIGH	$\text{sqrt}(1 - \text{raw_scalar})$ is inverted: maximum-confidence signals produce zero position size. Likely a bug that coincidentally dampens all trades.
Rolling z-score warm-up	MEDIUM	$\text{min_periods}=1$ produces unstable z-scores during the first ~252 days, potentially generating spurious signals.
PriceExp in option chain	LOW	Expiry spot price is present in the chain data at entry time. Used only for P&L; settlement (not entry decisions), but should be isolated.

Detailed Analysis

Issue 1: XGBoost Model Training Window (CRITICAL)

The VRP signal consumes RV forecasts from an XGBoost model via the column 'vrp_xgb'. The critical question is: **what was the training set for this model?**

If the model was trained on data that includes 2020 (or any period after the backtest start date), the model learns the statistical signature of extreme volatility regimes. During the backtest, when it encounters features in late January or February 2020 that resemble pre-crash conditions, it predicts elevated RV. This collapses $VRP = VIX - RV_{pred}$ towards zero or negative, blocking trade entry. The strategy appears prescient, but it is simply pattern-matching on future data.

There is nothing in the uploaded code that enforces an expanding-window or walk-forward retraining protocol. The DataManager loads pre-computed forecasts from a file, suggesting the model was trained once on the full dataset.

Fix: Implement strict walk-forward validation. At each rebalance date t , the XGBoost model must be trained only on data up to $t - h$ (where h is the forecast horizon). Retrain periodically (e.g. quarterly) with an expanding window. Store timestamped model artefacts to ensure reproducibility.

Issue 2: Prediction Interval Calibration (CRITICAL)

The regime signal (Signal 2) uses prediction interval widths from a VIX regression model. If this model was fit on the full sample, the residual distribution used to construct the 95% PIs includes COVID-era residuals. This inflates the residual variance estimate, which paradoxically could make pre-COVID PIs appear narrower relative to this inflated baseline (or, if the model learns regime-conditional heteroskedasticity, it widens PIs precisely when features resemble pre-crash conditions).

Fix: PIs must be constructed using only data available up to each prediction date. Use a rolling or expanding residual window for PI calibration.

Issue 3: Supplementary Classifier Leakage (CRITICAL)

The composite signal applies reductions based on VIX and RV classification models (pred_class_optimal, pred_prob). These are binary classifiers predicting whether VIX/RV will rise. If trained on the full sample, they learn that certain feature combinations precede large VIX spikes, and would flag the pre-COVID period with high probability, reducing the scalar and blocking or shrinking trades.

Fix: Same walk-forward protocol as Issue 1. Classification probabilities must be genuinely out-of-sample.

Issue 4: Composite Scalar Inversion (HIGH)

In composite.py, the final position scalar is computed as:

$$\text{final_scalar} = \sqrt{1.0 - \text{raw_scalar}} \text{ where } \text{raw_scalar} = S1 * S2$$

This is monotonically decreasing in raw_scalar. When both signals are maximally favourable (raw_scalar = 1.0), the output is 0.0: no trade at all. When signals are weak (raw_scalar near 0), the output is near 1.0: maximum position. This is the opposite of the intended behaviour and means the strategy systematically takes its largest positions when conditions are worst, and its smallest when conditions are best.

Likely intended formula: $\text{final_scalar} = \sqrt{\text{raw_scalar}}$, which would dampen the multiplicative interaction while preserving the correct monotonicity.

Issue 5: Rolling Z-Score Warm-Up (MEDIUM)

Both signals use `min_periods=1` in the rolling z-score computation. During the first 252 days of the backtest, the rolling mean and standard deviation are computed from progressively fewer observations. With, say, 10 observations, a single outlier can produce a z-score of +3 or -3, generating spurious EXTREMELY_RICH or INVERTED regimes.

The VRP signal partially addresses this with an `'is_warmup'` flag that halves the confidence score, but this flag is never actually used in trade decisions. The regime signal has no warm-up handling at all.

Fix: Either (a) set `min_periods=252` and begin the backtest 252 days after the first data point, or (b) use an expanding window with a minimum of 63 observations (one quarter) before producing actionable signals.

Issue 6: PriceExp Column in Option Chain (LOW)

The `trade_builder` reads `'PriceExp'` from the option chain at entry time to populate `spot_at_expiry`. This column contains the underlying price at expiration, which is future information at the time of trade entry. While it is used only for P&L computation (not trade selection), its presence in the entry-time DataFrame is poor practice and creates a risk that it could accidentally leak into feature engineering or entry logic.

Fix: Remove `PriceExp` from the entry chain. Look up expiry-date spot separately at settlement time using the market data source.

Part 4: Recommended Fixes

Priority 1: Walk-Forward Model Training

This is the single most important fix. All three ML components (XGBoost RV regression, VIX regression for PIs, and the supplementary classifiers) must be retrained using only data available up to the prediction date. A practical implementation:

1. Define a minimum training window (e.g. 756 days / 3 years).
2. At each quarterly rebalance date, retrain on data from [start, t - h].
3. Generate out-of-sample predictions for [t - h + 1, t_next_rebalance].
4. Stitch the out-of-sample predictions into a single time series for the backtest engine.

This will definitively resolve whether COVID avoidance was genuine signal or leakage. Expect Sharpe ratio to decrease and drawdowns to increase relative to the current results.

Priority 2: Fix the Composite Scalar

Replace the formula in composite.py:

```
# Current (inverted):  
final_scalar = math.sqrt(1.0 - raw_scalar)  
  
# Corrected:  
final_scalar = math.sqrt(raw_scalar)
```

Priority 3: Z-Score Warm-Up Guard

In both `vrp_signal.py` and `regime_signal.py`, change `min_periods` from 1 to at least 63 (one quarter). Add a guard in the `compute()` method to return `None` during the warm-up period, which will cause the composite signal to produce FLAT (no trade).

Priority 4: Isolate Expiry Spot Data

Remove `PriceExp` from the option chain `DataFrame` passed to `build_put_spread()`. Instead, have the engine look up the spot price at each trade's expiry date during the settlement phase, using a separate market data query.

If the strategy still avoids COVID after implementing walk-forward training, that is strong evidence of genuine predictive power. Until then, the COVID avoidance should be treated as a look-ahead artefact.